MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

Practice 1 | Version 202610122020.40 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

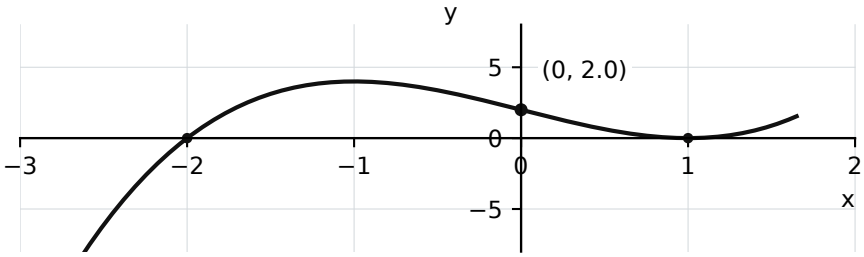
QUESTIONS	ANSWERS
1. Find all zeros: (a) list Rational Root Theorem candidates, then test only 2 and -2; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros. $p(x) = x^3 - x^2 - 3x + 2$	1. (a) Candidates/root — 5 pts (b) Quotient — 7 pts (c) Other zeros — 8 pts
2. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts. $f(x) = \frac{(x-1)(x+3)}{(x-5)(x-1)}$	2. (a) Domain — 4 pts (b) Hole — 4 pts (b) Vertical — 3 pts (b) Horizontal — 3 pts (c) x-intercept — 3 pts (c) y-intercept — 3 pts
3. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. $f(x) = \sqrt{x+1} - 2$	3. (a) Domain — 4 pts (a) Range — 4 pts (b) x-intercept — 3 pts (b) y-intercept — 3 pts (c) Infinite end — 4 pts (c) Finite end — 2 pts

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QUESTIONS	ANSWERS
4. Find (a) the vertex and the minimum or maximum value (state which); (b) all zeros. $f(x) = (x+1)^2 - 4$	4. (a) Vertex 6 pts (a) Minimum 4 pts (b) Zeros 10 pts
5. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. 	5. Polynomial 16 pts Left end 2 pts Right end 2 pts

Instructor answer key: MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a) Candidates/root
$\{-2, -1, 1, 2\};\ x = 2$ — 5 pts

(b) Quotient
$x^2 + x - 1;\ R = 0$ — 7 pts

(c) Other zeros
$x = -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$ — 8 pts

2. (a) Domain
$\mathbb{R} \setminus \{1, 5\}$ — 4 pts

(b) Hole
$(1,\ -1)$ — 4 pts

(b) Vertical
$x = 5$ — 3 pts

(b) Horizontal
$y = 1$ — 3 pts

(c) x-intercept
$(-3,\ 0)$ — 3 pts

(c) y-intercept
$(0,\ -\frac{3}{5})$ — 3 pts

3. (a) Domain
$[-1, \infty)$ — 4 pts

(a) Range
$[-2, \infty)$ — 4 pts

(b) x-intercept
$(3,\ 0)$ — 3 pts

(b) y-intercept
$(0,\ -1)$ — 3 pts

(c) Infinite end
$x \to \infty :\ f(x) \to \infty$ — 4 pts

(c) Finite end
$x \to -1^{+} :\ f(x) \to -2$ — 2 pts

TEST PAGE 2

4. (a) Vertex
$(-1,\ -4)$ — 6 pts

(a) Minimum
-4 — 4 pts

(b) Zeros
$x = -3,\ 1$ — 10 pts

5. Polynomial
$1(x + 2)(x - 1)^2$ — 16 pts

Left end
$x \to -\infty :\ f(x) \to -\infty$ — 2 pts

Right end
$x \to \infty :\ f(x) \to \infty$ — 2 pts